First Half

	Financial calculation	Excel syntax
Annuity	Present value of annuity including a final payment of FV	=pv(rate, nper, pmt, [fv], [type])
	Payment that makes present value equal to target value	=pmt(rate, nper, pv, [fv], [type])
	Number of payments needed to match present value to target value	=nper(rate, pmt, pv, [fv], [type])
	Interest rate per period that matches present value and target value	=rate(nper, pmt, pv, [fv], [type])
Other	Final payment that matches present value and target value	=fv(rate, nper, pmt, [fv], [type])
	Present value of a range of cash flows starting in one period	=npv(rate, range of values)
	Rate of return that makes the present value of a stream of payments zero (the first payment is immediate)	=irr(range of values, [guess])

=normsinv() Inverse standard normal CDF function
=normsdist() Standard normal CDF function

$PMT_0 = \frac{PMT}{1+g}$ $\quad P_{t+j} - \mu = (P_t - \mu)(1-\kappa)^j$ $\quad EAY = \left(\frac{100}{Price}\right)^{365/days} - 1$

$r^* = \frac{r-g}{1+g}$ $\quad (1+EAR) = (1+\frac{r}{k})^k$ $\quad PV = \frac{f_{t-1,t}F}{(1+r_{st})^t}$

No Arbitrage: f is set to ensure (a) and (b) have the same FV

$t = -\ln(1-u)/$ lambda

$1.022^2 = 1.02(1+f)$ ➔ f = 2.4004%

Discrete time discounting

$$\rho = \frac{r-i}{1+i}$$

Continuous discounting

$$\rho = r - i$$

$$\frac{F(t)}{X_{CNY}} = \frac{(1+r_{CNY})^t}{(1+r_{USD})^t} \equiv (1+\delta)^t$$

$E[f(z)] = \bar{X} \pm 2s/\sqrt{N}$

Bond (promised) yield = $r + k + \pi\lambda$
Credit spread = $k + \pi\lambda$

$z_2 = \rho z_1 + z_3\sqrt{1-\rho^2}$

$V_0 = \frac{C}{r}\left(1 - \frac{1}{(1+r)^n}\right)$ *for* $r \neq 0$

$PVLGP(a,b,r,s) = \frac{\left(\frac{a+bs}{r} + \frac{b}{r^2}\right)}{(1+r)^{s-1}}$

Sharpe Ratio = $\frac{r_i - r}{\sigma_i} = \frac{Expected\ dollar\ return - r.f.funding\ costs}{Dollar\ standard\ deviation}$

ABM

Cov

GBM

$E[$

$Cov[$

S_t

$S_{t+j} = \mu + ($

MR

$E[S_{t+}$

$Var[S$

$Cov[S_{t+j}, S_{t+m}|S_t] =$

$A = 1'\Sigma^{-1}1$
$B = 1'\Sigma^{-1}r$
$C = r'\Sigma^{-1}r$
$D = AC - B^2$

$\sigma^2(R) = \frac{AR^2 - 2B}{D}$

$s^2(X_h'b) = s^2X_h'(X'X)$

Second Half

Table 1
Derivation of the Equilibrium Valuation of n Correlated Cash Flows in the 1-period CFPT Model

	Variance Utility	Standard Deviation Utility
Expected wealth	$N'\mu + (W_0 - N'V)(1+r)$	
Variance of wealth	$N'\Sigma N$	
Expected utility	$N'\mu + (W_0 - N'V)(1+r) - \frac{1}{2}AN'\Sigma N$	$N'\mu + (W_0 - N'V)(1+r) - A(N'\Sigma N)^{1/2}$
First order condition	$\mu - V(1+r) - A\Sigma N = 0$	$\mu - V(1+r) - A\Sigma N/(N'\Sigma N)^{1/2} = 0$
Valuation with market clearing (N=1)	$V = \frac{\mu - A\Sigma 1}{(1+r)}$	$V = \frac{\mu - A\Sigma 1/(1'\Sigma 1)^{1/2}}{(1+r)}$
GVE	$rV + A\Sigma 1 = \mu - V$	$rV + \frac{A\Sigma 1}{(1'\Sigma 1)^{1/2}} = \mu - V$
Aggregate A	$A = \frac{1'(\mu - V(1+r))}{1'\Sigma 1}$ "Market price of variance risk"	$A = \frac{1'(\mu - V(1+r))}{(1'\Sigma 1)^{1/2}}$ "Dollarized Market Sharpe Ratio"

$$V_C = \left(V - \frac{X}{1+r}\right)N(d^*) + \frac{\sigma n(d^*)}{1+r} = \frac{\mu_c^*}{1+r}$$

$$d^* = \frac{(1+r)V - X}{\sigma}$$

$$C_0 = S_0 N(d_1) - Xe^{-rT}N(d_2)$$

$$d_1 = \frac{\ln\left(\frac{S_0}{X}\right) + (r + 0.5\sigma^2)T}{\sigma\sqrt{T}} \qquad d_2 = d_1 - \sigma\sqrt{T}$$

$$F_{0,T} = S_0 \exp[(r-\delta)T]$$

Ta
Risk Premium Calculations in the

Calculation	Dollars
Total expected return	$\mu - V$
Risk premium	$\mu - (1+r)V = A$
Covariance with market	$\Sigma 1$
Variance of market	$1'\Sigma 1$
Beta	$\frac{\Sigma 1}{1'\Sigma 1}$
Risk premium⊘Beta (constant for all securities)	$A1'\Sigma 1$

*The symbol "⊘" is being used to represent the Hadamard qu quotient of two vectors or matrices

$$r_m = \frac{1'(\mu - V)}{1'V}$$

r_i

$r_i - r$

- Partition the cash flows into known [1..j] a time j
- Then we have $C \equiv \begin{pmatrix} C_j \\ C_N \end{pmatrix}$ $\mu \equiv \begin{pmatrix} \mu_j \\ \mu_N \end{pmatrix}$
- This gives us a kind of multiple regression decomposition of the covariance matrix
  - $E(C_N|C_j) = \mu_N + \Sigma_{Nj}\Sigma_{jj}^{-1}(C_j - \mu_j)$
  - $\Sigma_j = \Sigma_{NN} - \Sigma_{Nj}\Sigma_{jj}^{-1}\Sigma_{jN}$ *for* $j = 1..n-1$ w
  - (with the single subscript, the Σ is the covari given the cash flows up to time j)

Variance formula

$V_j = B'E[C_N|C_j] - kB'(1'\Sigma_j 1$

Standard deviation formula

$V_j = B'E[C_N|C_j] - kB'(sqrt($